

SUMMER AI WORKSHOP

How LLMs Work

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Business + Technology @





LLM MODEL COMPONENTS: A TWO-PART VIEW

LLM MODEL

PART 1: RAW NEXT-TOKEN PREDICTION



PRE-TRAINING (BASE MODEL)



Base Model

PART 2: TRAINING/TUNING TO FOLLOW INSTRUCTIONS & BE SAFE



FOLLOW INSTRUCTIONS

e.g., "Summarize this text"



AVOID SOME HARMFUL OUTPUTS

e.g., Refuse hate speech



USE CONVERSATIONAL PATTERNS

e.g., Acknowledge, use common phrasing



PREFER ANSWERS HUMANS RATE AS HELPFUL

e.g., RLHF: Reward modeling, human preference



- Supervised Fine-Tuning (SFT)
- Reinforcement Learning from Human Feedback (RLHF)



ALGORITHM > SYNTACTIC > HUMAN > EXECUTABLE
BLOCKS EXPRESSION LOGIC

AI

judgment

prediction

planning

perception

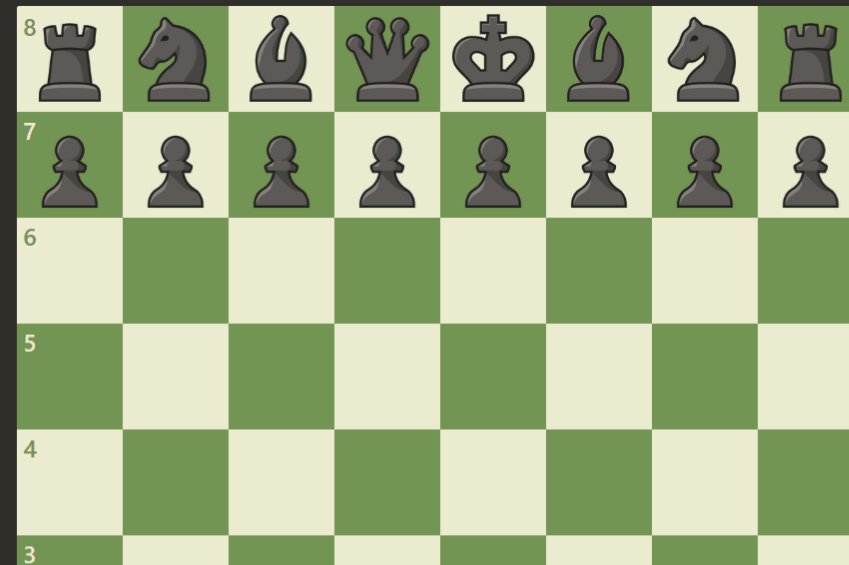
classification

language

AI

Sent Ma

Spam (3



25

Maximum 3200

Play Bots



Coach 5 bots

Adaptive 5 bots

Beginner 15 bots

Intermediate 15 bots

Advanced 20 bots

Master 10 bots

Athletes 11 bots

07:16

4G



860 m

D837:
Chartres-de-
Bretagne

D837

D837

80

63
km/h

Kertrucks
- Renault

EQIOM Bétons
- Ouest

Suez - Sita
Ouest

D34

07:34
arrivée

18
min

11
km

VZW Wi-Fi



Face ID

AI

└─ Generative AI

text
images
music
code

video
slides
voices
designs

AI

└─ Generative AI

└─ Large Language Models

text

code (text)

math (text)

notation (text)

tables (text)

instructions

structured data

(text)

MODEL V INTERFACE

TOKENS

text:

The student submitted the assignment late.

tokens:

"The", " student", " submitted", " the", " assignment", " late", "."

text:

antidisestablishmentarianism

tokens:

"anti", "dis", "establishment", "arian", "ism"

TOKENS

****One token at a time****

****Tokenization affects cost and limits****

****Weird text confuses models****

****It does not see text like we do****

TOKENS

text:

AI can help, but it can also be wrong.

tokens:

"AI", " can", " help", ",", " but", " it", " can", " also", " be", " wrong", "."
(11)

text:

Please explain this like I'm a beginner.

tokens:

"Please", " explain", " this", " like", " I", "'m", " a", " beginner", "." (9)

TOKENS

emoji:



tokens:

"  " (1)

emoji:



tokens:

"  ", "  " (2)

TOKENS

emoji:



tokens:

"", "", " " (3)

name:

Paul Wayland

tokens:

"Paul", " Way", "land" (3) or "Paul", "Wayland" (2)

TOKENS

name:

Inter American University of Puerto Rico

tokens:

"Inter", " American", " University", " of", " Puerto", " Rico" (6)

name:

García-Márquez

tokens:

"Garc", "ía", "-", "M", "ár", "quez" (6)

“apple”

[0.0182, -0.0127, 0.0441, 0.0038, -0.0295, 0.0614,
-0.0079, -0.0352, 0.0206, 0.0143, -0.0528, 0.0091, ...]

given previous tokens,
predict the next token

given previous tokens,
predict the next token

"Puerto Rico is located in the"

" Caribbean" (0.58)

" northeastern" (0.12)

" Greater" (0.07)

" United" (0.01)

other tokens (0.18)

Temperature	Behavior
Low	More focused, consistent, deterministic
Medium	Balanced, useful for normal work
High	More creative, more surprising, more likely to wander

NOT **retrieved** but ****BUILT****

system instructions

+ current prompt

+ answer so far



MODELS FORGET BECAUSE

1. the conversation got too long, and older material was dropped, summarized, or compressed.
2. long contexts create attention problems. Important details can get buried.
3. memory may store selected facts, preferences, or summaries, not every exact prior exchange

Type	What Happens
Context overflow	Old content falls outside the active window
Attention dilution	Important details remain present but get ignored
Summary loss	Earlier details are compressed and nuance disappears
Product memory gap	The app remembers some facts, not the full conversation
User assumption error	The user assumes the model saw a file/page/message it did not see

The model can only reason from what is in front of it, plus patterns learned during training.
If the key detail is missing, buried, or ambiguous, the answer may drift.

Strong prompts restate essentials: goal, context, constraints, data, and desired output.

hallucination ≠ **lying**

LARGE *language* MODEL

hallucination triggers

1. Asking for obscure facts.
2. Asking for citations without giving sources.
3. Asking about recent events without web access.
4. Asking for local or institutional details.
5. Asking the model to fill gaps instead of admitting uncertainty.
6. Overly broad prompts like “tell me everything about...”
7. Pressure prompts like “don’t say you don’t know.”

GOOD PROMPTS

Use only the sources I provide. If the answer is not in the sources, say “not found in provided sources.”

GOOD PROMPTS

Separate confirmed facts from inferences and uncertainties.

****LLMs are confidence machines, not truth machines.****

They can be truthful, useful, and often brilliant,
but truth is not guaranteed by fluency.

GOOD PROMPTS

Answer in three sections: confirmed facts,
likely inferences, and unknowns.
Do not invent missing information.

GOOD PROMPTS

Give me at least two reliable sources.